Run report: run-20261015T0200Z (04b genus-3 IC-2LP rebalance)

Descriptive record of what ran. Interpretation is in the study README and REPORT.pdf.

Spec and commit. Protocol: `research/ic_improve_20261014/04b-genus3-2lp-rebalance/PROTOCOL.md`, frozen at commit 9752513ea (sha256 633e69bb5f71…). κ choice frozen at 288e74293 (`tuning/kappa_choice.json`). Tool: `examples/ic_improve_04b_genus3_2lp.rs` (sha256 595e3efa…), built with `cargo build --release --example ic_improve_04b_genus3_2lp` (binary sha256 8077f4d8…), rustc 1.97.0. Analysis: `examples/ic_improve_04b_analyze.rs`.

Host and isolation. 4 vCPU Intel Xeon @ 2.80 GHz, 15 GiB, Linux 6.18; shared cloud container, **no isolation earned**. Four jobs in parallel under `timeout 3600` and `prlimit --as=3 GiB` (`run_jobs.sh`). Start 2026-10-15T01:55Z; test records finished 04:39Z (`results/env-run-20261015T0200Z.txt`).

Arms, seeds, workloads. Seed 20261016. Sizes $k = 10..20$, q = largest prime below 2^k. 7 GHS genus-3 curves per size (`results/curves/`). Tuning: $k = 14..20$ × κ ∈ {0.5, 0.7, 1, 1.4, 2} × curves 5–7 × targets 1–2. Test: curves 1–4 × targets 1–3 at κ = 2 for every k (BL and A2 from the same relation phase) and at κ = 1.4 for k = 17–20 (A1; κ* = 2 at 14–16 reuses the κ = 2 run). Rho: $k = 10..14$, curves 1–4, target 1.

job set	requested	completed	failed	records
curves	11	11	0	`results/curves/` (77 curves)
tuning solves	210	210	0	`results/tuning/`
test solves	180	180	0	`results/test/`
rho	20	20	0	`results/rho/`
orchestration error	—	—	1	`results/orchestration_errors.jsonl`: an empty job list launched once (no tool arguments, exit 2); not a test job

Measured values (unit: multiplications in $\mathbb{F}_q$ plus mod ℓ; 12 test instances per size; 95% bootstrap intervals in `analysis.json/tables.md`). Every one of the 360 test logarithms (180 solves × 2 linear-algebra routes), every tuning logarithm and every rho logarithm satisfied $[\log]P = Q$ and equalled the planted value. Peak RSS 119 MB; longest job 490 s. One test solve replayed after the run reproduced its record's counts exactly.

k	BL mean total	A1 mean total	A2 mean total	BL rows / nnz per row	rho mean S (n = 4)
10	3.142e6	—	2.559e6	213 / 4.92	550.0
12	1.560e7	—	1.199e7	522 / 6.26	439.7
14	9.686e7	9.686e7	7.566e7	1301 / 7.76	571.4
16	6.731e8	6.731e8	5.575e8	3261 / 9.55	—
18	4.387e9	4.301e9	3.801e9	8202 / 11.71	—
20	3.049e10	2.930e10	2.763e10	20653 / 14.38	—

Even sizes shown; all sizes are in tables.md. Derived (analysis tool, OLS over $k = 14..20$): BL total exponent 1.383, A1 1.373, A2 1.419; BL relation 1.332, linear algebra 1.447.

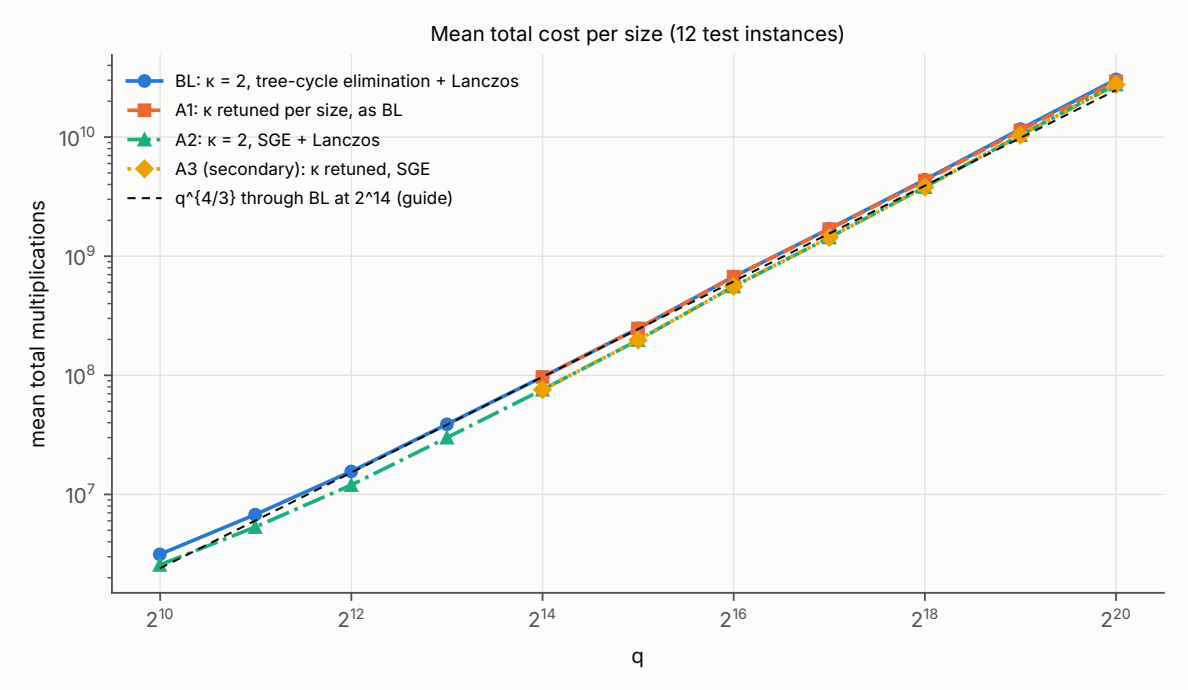


Figure 1: Recorded mean total multiplications per size, 12 test instances per arm, run-20261015T0200Z. The dashed line is a $q^{4/3}$ guide through BL at 2^{14}.

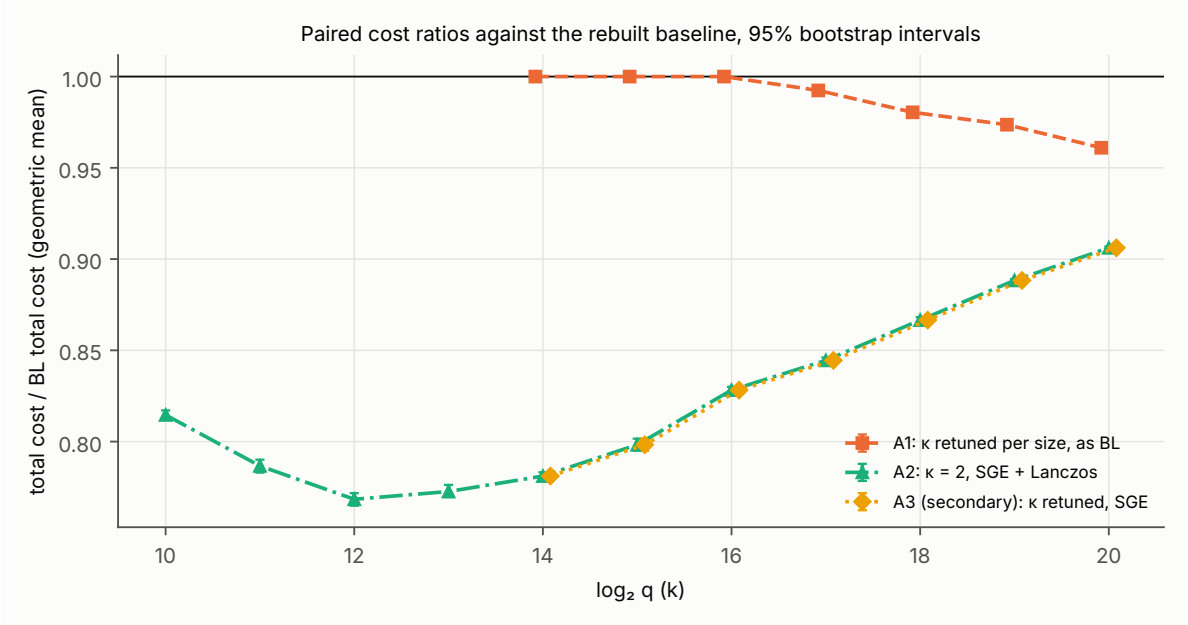


Figure 2: Recorded paired ratios arm/BL (geometric mean, 95% bootstrap intervals), run-20261015T0200Z.

Records. `results/curves/curves_k*.jsonl`, `results/tuning/tu_*.jsonl`, `results/test/te_*.jsonl`, `results/rho/rho_*.jsonl`; analysis.json and tables.md from `ic_improve_04b_analyze report`. Missing data: none.